

raw data

1. Data Layer

data_loader.py

Vrbancic et al. (2020)
88,647 x 111 | 34.57% phishing | 1:1.89

URL-Phish (2025)
116,600 x 22 | 14.24% phishing | 1:6.02

2. Preprocessing

preprocessing.py

Structural cleaning
constant / duplicate cols

Stratified subsample
20,000 rows; ratio kept

Stratified 80/20 split
test set sealed

training set only

3. Imbalance Treatment

imbalance.py

Data-level
Random Over / Under
SMOTE, ADASYN

Hybrid
SMOTEENN
SMOTETomek

Algorithm-level
Cost-sensitive
learning

treated training data

4. Model Training

models.py

imblearn Pipeline: scaler -> sampler -> classifier, inside GridSearchCV (stratified 5-fold)

Decision Tree

Random Forest

Support Vector Machine

fitted models

5. Evaluation and Analysis

Performance metrics
Precision, Recall, F1, ROC-AUC, PR-AUC, MCC
evaluation.py

Comparative analysis
aggregation and result tables
analysis.py

Significance testing
Friedman, Wilcoxon, McNemar
statistical_tests.py

Explainability
SHAP global and local attribution
explainability.py

Cross-cutting: reproducibility and orchestration

Fixed seeds (42, 1, 2) propagate to subsampling, splitting, folds, samplers and classifiers
config.py | *experiment.py* | *run_pipeline.py*